Maijunxian Wang

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Education

University of California, Berkeley Sep 2025 - Spring 2027 (expected)
BA in Cognitive Science GPA: **3.965/4.0**

University of California, Davis Sep 2023 - Jun 2025
Undergraduate coursework in Cognitive Science; transferred to UC Berkeley GPA: **3.981/4.0**

Research Interests

Visual reasoning; video generation and world models; cognitively grounded AI.

I study vision as a reasoning medium distinct from language, combining cognitively grounded task design, scalable data construction, and video-model post-training to investigate how models infer, transform, and predict world states through visual representations.

Honors

Sayan Gul Award - Best Undergraduate Student Paper CogSci 2026
Cognitive Science Society, for *Egocentric Bias in Vision-Language Models*.

Research Experience

Hippocampal May 2026 - Present

Omni-Reason
Project Lead and First Author; research in progress

- Led video-model post-training, designing data mixtures and experiments and analyzing results; tuned task sampling, learning rates, and training duration to produce a single model capable of performing **100 computer vision tasks**.
- Designed and built an agent-assisted data-production platform with human quality review and PR tracking; led coordination of **20+ research contributors** to construct **24M input-target video pairs across 300 tasks**.

Training Object Permanence in World Models
Preprint, 2026

- Converted **100 existing Blender tasks and scenes** into parameterized, scalable generators, supporting production of the project's **1.5M training samples** for studying object permanence in video models.

VBVR Research / Video-Reason Dec 2025 - Present

A Very Big Video Reasoning Suite (VBVR)
ICML 2026

- Designed the full suite of visual reasoning tasks spanning perception, transformation, spatial reasoning, abstraction, and knowledge, translating cognitive abilities into generative task specifications.
- Generated and rendered task data for a video-reasoning resource with **1M training examples**, supporting model training and systematic evaluation.

VBVR-Pro: A Scalable and Verifiable Suite for Native Visual Reasoning
Preprint, 2026

- Designed **150 new visual reasoning tasks** beyond the original VBVR suite, expanding its coverage to **300 tasks**.

Demystifying Video Reasoning
ECCV 2026

- Analyzed layer-wise representations across **40 DiT blocks**, identifying spatially sparse token activations and a shift from global structure toward task-relevant objects.

Research Experience (continued)

Grow AI Like a Child Feb 2025 - Nov 2025

Egocentric Bias in Vision-Language Models (FlipSet)
CogSci 2026

- Led study design, evaluation of **103 vision-language models**, analysis, and writing; developed controlled visual perspective-taking tasks.
- Designed controls separating perspective judgment from mental rotation, revealing systematic egocentric bias and failures to compose component abilities.

Assessing Perceptual Metacognition in Vision-Language Models
CogSci 2026

- Scaled numerosity, orientation, and brightness tasks into experimental stimuli; evaluated models and analyzed confidence, calibration, and metacognitive measures to compare implicit confidence with explicit self-reports.

Vision Language Models Cannot Reason About Physical Transformation
ConservationBench; ICML 2026

- Designed and filmed conservation and non-conservation tasks across number, length, size, and volume; computed and visualized performance correlations to investigate physical-transformation reasoning and biases.

Publications

* Equal contribution. Long author lists are abbreviated; author order is preserved.

Conference Papers

1. **Maijunxian Wang***, R. Wang*, J. Lin*, R. Ji*, et al.
A Very Big Video Reasoning Suite. ICML 2026. [Paper]
2. **Maijunxian Wang***, Y. Li*, B. Wang, T. Zhao, R. Ji, Q. Gao, E. Liu, H. Deng, D. Luo.
Egocentric Bias in Vision-Language Models. CogSci 2026. *Sayan Gul Award.* [Paper]
3. R. Ji*, **Maijunxian Wang***, Q. Gao, Y. Li, H. Deng, D. Luo.
Assessing Perceptual Metacognition in Vision-Language Models. CogSci 2026. [Paper]
4. R. Wang, Z. Cai, F. Pu, J. Xu, W. Yin, **Maijunxian Wang**, et al.
Demystifying Video Reasoning. ECCV 2026. [Paper]
5. D. Luo*, Y. Li*, **Maijunxian Wang**, et al.
Vision Language Models Cannot Reason About Physical Transformation. ICML 2026. [Paper]

Preprints

6. J. Xu*, R. Wang*, F. Pu*, **Maijunxian Wang***, R. Ji*, T. Zhou*, et al.
VBVR-Pro: A Scalable and Verifiable Suite for Native Visual Reasoning. 2026. [Paper]
7. H. Zhang, F. Yu, D. Luo, . . . , **Maijunxian Wang**, . . . , H. Deng.
Training Object Permanence in World Models. 2026. [Paper]

Workshop Papers

8. B. Wang*, Y. Li*, Y. Qiao, **Maijunxian Wang**, et al.
Increasing Computation Resolves Conflicts in Vision Language Models. MMRAgI Workshop at CVPR 2026. [Paper]
9. H. Sun, B. Wang, S. Yu, . . . , **Maijunxian Wang**, D. Luo, H. Deng.
Probing Perceptual Constancy in Large Vision-Language Models. ES-Reasoning Workshop at ICLR 2026. [Paper]